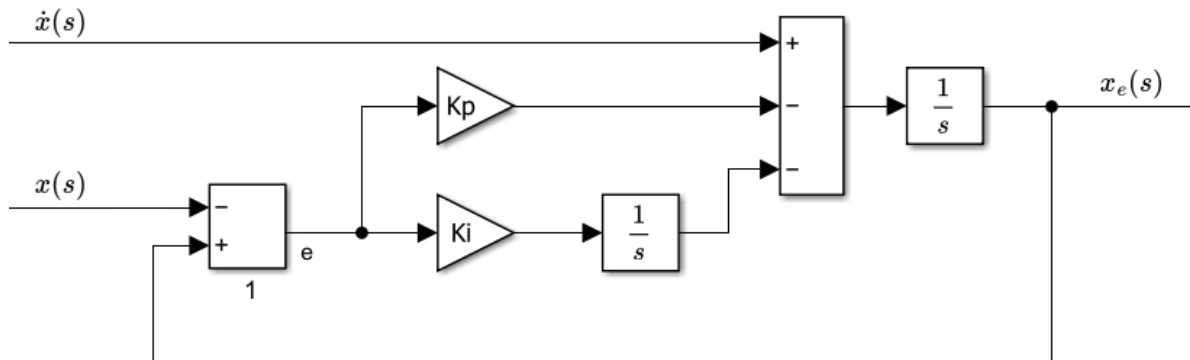


Recommended Assessment

Sensor Fusion

1. What do you observe about the forward speed estimate based on the accelerometer reading when the QBot is stationary? Include a screenshot to show your observations. Why does this happen?
2. What do you observe about the forward speed estimate based on the wheel speeds? How does this estimate compare to the accelerometer-based estimate? Include a screenshot to show your observations.
3. Explain what happened when we added proportional control with gain K_p to estimate forward speed. Why does this happen? How does this new estimate compare to the estimates from separate sensors?
4. Explain why the K_i gain and integrator was added.
5. Explain how tuning the gains K_p and K_i affect the characteristics of the complementary filter.

6. Derive the transfer function for the block diagram and show that it is a valid implementation of a complementary filter.



7. Compare the complementary filter and EKF implementations to estimate position. Include a screenshot to show your observations. When would you choose one over the other?